Rational 5-torsion and the 2-rank of geometrically split Jacobians

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Theorem. *Let $A/\mathbb{Q}$ be a principally polarized abelian surface that is simple over $\mathbb{Q}$ and geometrically split. If $A(\mathbb{Q})$ has a point of order 5, then $A(\mathbb{Q})[2] \cong (\mathbb{Z}/2)^r$ with $r \leq 2$.*

Corollary. *No genus-2 Jacobian over $\mathbb{Q}$ that is geometrically split and simple over $\mathbb{Q}$ has a rational torsion subgroup containing $(\mathbb{Z}/2)^3 \times \mathbb{Z}/5$. In particular $[2,2,10]$, $[2,2,20]$ and $[2,2,2,10]$ do not occur as the rational torsion subgroup of such a Jacobian. Together with the 2-rank obstruction for Jacobians split over $\mathbb{Q}$, no geometrically split genus-2 Jacobian over $\mathbb{Q}$ has rational torsion subgroup $[2,2,2,10]$.*

Here "geometrically split" means that $A_{\bar{\mathbb{Q}}}$ is isogenous to a product of two elliptic curves. We impose no restriction on the degree of this isogeny, on the field of definition of the geometric endomorphisms or on the integral endomorphism ring. The principal polarization is used in one place only, Lemma 4. Along the way we prove a statement that needs neither a point of order 5 nor a polarization:

Proposition A. Let $A/\mathbb{Q}$ be an abelian surface such that $A_{\bar{\mathbb{Q}}}$ is isogenous to the square of an elliptic curve without complex multiplication, and such that the field of definition of the geometric endomorphisms of A is a cyclic quartic extension of $\mathbb{Q}$. Then $A(\mathbb{Q})[2^\infty]$ has order at most 4.

Outline. We follow the case distinction of the proof for $[2,2,14]$. The CM case follows from a point count at 3: no abelian surface over $\mathbb{F}_3$ has exactly 40 points (Section 2). If $A_{\bar{\mathbb{Q}}}$ is the square of a curve without CM, the residual-representation lemmas of the proof for order 63, applied at the prime 5, leave only the endomorphism fields that are cyclic of degree 2 or 4 (Section 3). In the cyclic quartic case the determinant character of order 4 forces $\#\widetilde{A}(\mathbb{F}_p) \not\equiv 0 \pmod 8$ at a positive density of primes, which gives Proposition A (Section 4). In the remaining cases A is isogenous to the Weil restriction of an elliptic curve E over a quadratic field K with $E^\sigma \neq E$. A principal polarization and rational 2-rank 3 produce such an E with full K-rational 2-torsion, and the point of order 5 then gives a K-point on the modular curve $X_1(2,10)$, an elliptic curve of rank 0 over $\mathbb{Q}$. At primes inert in K, $\#\widetilde{A}(\mathbb{F}_p) = \#\widetilde{E}(\mathbb{F}_{p^2})$ must be divisible by 8. By Chebotarev's theorem this forces certain norms to be rational squares. A quadratic point with rational x-coordinate comes from an elliptic curve over $\mathbb{Q}$, which contradicts simplicity over $\mathbb{Q}$; at every other quadratic point that is not a cusp, these norms are not all squares in the required way (Section 5).

1 Conventions and inputs

All number fields are subfields of a fixed algebraic closure $\bar{\mathbb{Q}}$. For a number field k we write $G_k = \mathrm{Gal}(\bar{\mathbb{Q}}/k)$. For a prime ℓ, χ_ℓ is the ℓ-adic cyclotomic character, T_ℓ is the Tate module and $V_\ell = T_\ell \otimes \mathbb{Q}_\ell$. For an abelian variety X, $\mathrm{End}^0 = \mathrm{End} \otimes \mathbb{Q}$ and $\mathrm{Hom}^0 = \mathrm{Hom} \otimes \mathbb{Q}$.

Conjugate varieties and morphisms are as in [P2214, Section 1]: for $g \in G_{\mathbb{Q}}$, Y^g is the conjugate of Y and $g : Y(\bar{\mathbb{Q}}) \to Y^g(\bar{\mathbb{Q}})$ the induced bijection. For a quadratic field K we write σ for its nontrivial automorphism, and $\tilde{\sigma} \in G_{\mathbb{Q}}$ for a lift of σ. We write $\mathbb{Q}^{\times 2}$ for the nonzero rational squares.

- **(I1) Reduction** [ST], as stated in [P63, Section 1] and [P2214, (I2)]: let K' be a finite extension of $\mathbb{Q}_p$ with residue field k, A an abelian variety over K' and $\ell \neq p$ a prime. (a) Good reduction is detected by the inertia action on $T_\ell(A)$ [ST, §1, Theorem 1 and Corollary 2]; in particular it is preserved by K'-isogenies. (b) If A has good reduction $\widetilde{A}$ and m is prime to p, then $A(K')[m]$ injects into $\widetilde{A}(k)$ [ST, Lemma 2], and $T_\ell(A) \cong T_\ell(\widetilde{A})$ compatibly with Frobenius. (c) If A has potentially good reduction, then it has good reduction over a finite extension of K' with residue field k [ST, §2]. (d) (Weil.) For an abelian variety B of dimension g over $\mathbb{F}_q$, $\#B(\mathbb{F}_q) = P(1)$, where P is the characteristic polynomial of the Frobenius endomorphism on $V_\ell(B)$, and the complex roots of P have absolute value $\sqrt{q}$; hence $\#B(\mathbb{F}_q) \leq (1+\sqrt{q})^{2g}$. By (b) and (d), if an abelian variety X over a number field k has good reduction $\widetilde{X}$ at a prime $\mathfrak{p} \nmid \ell$, then $\#\widetilde{X}(O_k/\mathfrak{p}) = \det\bigl(1 - \mathrm{Frob}_{\mathfrak{p}} \mid V_\ell(X)\bigr)$ for an arithmetic Frobenius element $\mathrm{Frob}_{\mathfrak{p}} \in G_k$ at $\mathfrak{p}$.

- **(I2) The case distinction** [P2214, (I3)]. A geometrically split abelian surface $A/\mathbb{Q}$ is of exactly one of the types **(N)** ($A_{\bar{\mathbb{Q}}}$ is isogenous to $E_1 \times E_2$ with E_1, E_2 not isogenous), **(CM)** ($A_{\bar{\mathbb{Q}}}$ is isogenous to E^2 with E a CM curve) and **(S)** ($A_{\bar{\mathbb{Q}}}$ is isogenous to E^2 with E without CM). In case (S), $D = \mathrm{End}^0(A_{\bar{\mathbb{Q}}}) \cong M_2(\mathbb{Q})$; L is the smallest field over which all geometric endomorphisms are defined; $\Gamma = \mathrm{Gal}(L/\mathbb{Q})$ embeds in $\mathrm{PGL}_2(\mathbb{Q})$, the element g acting on D by conjugation with $R_g \in D^\times$; and $\mathrm{End}^0_{\mathbb{Q}}(A) = D^\Gamma$. The group Γ is one of C_1, C_2, C_3, C_4, C_6, V_4, D_6, D_8, D_{12}, and $\Gamma \neq C_1$ if A is simple over $\mathbb{Q}$ [P63, Section 5]. An element $r \in \Gamma$ of order $n \in \{3,4,6\}$ has a unique representative R with $\mathrm{tr}\, R = 1$, and $R^2 - R + 1/c = 0$ with $c = 1, 2, 3$ respectively.

- **(I3)** [P63, Lemma 2] Let X be an abelian surface over a number field k, $E \subset X$ an elliptic curve defined over k, and $Q \in X(k)$ of prime order q. Then either $Q \in E(k)$, or the image of Q in $(X/E)(k)$ has order q.

- **(I4)** [P63, Lemma 3] In case (S), every elliptic curve in $A_{\bar{\mathbb{Q}}}$ is defined over L, and any two of them are L-isogenous.

- **(I5) Residual representations** [P63, Section 6]. Let p be an odd prime and A of type (S). Lemma 5 there (a polarization of A over $\mathbb{Q}$, of any degree) is used by Lemma 6 (*unramified order*) and Lemma 7 (*split blocks*); both conclude $\mathbb{Q}(\zeta_p) \subseteq L$ when $A(\mathbb{Q})[p] \neq 0$ and a Γ-stable $\mathbb{Z}_p$-order $\mathcal{B} \subset D \otimes \mathbb{Q}_p$ of the stated kind exists.

- **(I6) The determinant** [P2214, Lemma 3]. Let A be of type (S) with $\Gamma = \langle r \rangle$ cyclic of order $n \in \{3,4,6\}$, and R as in (I2). Then $F = \mathbb{Q}[R]$ is a field, because $X^2 - X + 1/c$ has negative discriminant ($F \cong \mathbb{Q}(i)$ for $n = 4$); as R is not a scalar, its centralizer in D is F, so $F = D^\Gamma = \mathrm{End}^0_{\mathbb{Q}}(A)$; and $N(R) = \det R = 1/c > 0$. The statement and proof of [P2214, Lemma 3] are written for $n \in \{3,6\}$, but they use only these facts and that the roots of unity in F have norm 1; they apply verbatim to $n = 4$, $c = 2$. For an elliptic curve $E \subset A_L$ and $H = \mathrm{Hom}^0(E_{\bar{\mathbb{Q}}}, A_{\bar{\mathbb{Q}}})$, a one-dimensional F-vector space, evaluation is an F-linear isomorphism $V_\ell(E) \otimes_{\mathbb{Q}} H \to V_\ell(A)$. Under it each $g \in G_{\mathbb{Q}}$ acts as $B_g \otimes \gamma_g$ with $B_g \in \mathrm{GL}(V_\ell(E))$ and $\gamma_g \in F^\times$, and

$$\varepsilon(g) := \gamma_g^2 / \deg(u_g)$$

for a quasi-isogeny $u_g : E \to E^g$ of degree $\deg(u_g) \in \mathbb{Q}_{>0}$ is a character $\varepsilon : G_{\mathbb{Q}} \to \mu(F)$ that factors through Γ, with $\varepsilon(r) = R/\bar{R}$ a primitive n-th root of unity. The $F \otimes \mathbb{Q}_\ell$-determinant of g is $\chi_\ell(g)\varepsilon(g)$.

- **(I7) Weil restriction** [Mi, §1]. Let K be a quadratic field and E an elliptic curve over K. The Weil restriction $\mathrm{Res}_{K/\mathbb{Q}} E$ is an abelian surface over $\mathbb{Q}$ with $(\mathrm{Res}_{K/\mathbb{Q}} E)(\mathbb{Q}) = E(K)$ and $(\mathrm{Res}_{K/\mathbb{Q}} E)_K \cong E \times E^\sigma$. Its $\bar{\mathbb{Q}}$-points are the pairs (x, y) with $x \in E(\bar{\mathbb{Q}})$ and $y \in E^\sigma(\bar{\mathbb{Q}})$; an element $g \in G_K$ maps (x, y) to (gx, gy), and an element $g \notin G_K$ maps it to (gy, gx). Accordingly $V_\ell(\mathrm{Res}_{K/\mathbb{Q}} E) \cong \mathrm{Ind}_{G_K}^{G_{\mathbb{Q}}} V_\ell(E)$. If E is the base change of an elliptic curve E_0 over $\mathbb{Q}$, then $\mathrm{Res}_{K/\mathbb{Q}} E$ is isogenous over $\mathbb{Q}$ to $E_0 \times E_0^{(K)}$, where $E_0^{(K)}$ is the quadratic twist.

- **(I8) Halving** [Kn, Theorem 4.2]. Let k be a field of characteristic different from 2 and 3, and $E : y^2 = (x - e_1)(x - e_2)(x - e_3)$ with distinct $e_i \in k$. A point $(x_0, y_0) \in E(k)$ lies in $2E(k)$ if and only if $x_0 - e_1$, $x_0 - e_2$ and $x_0 - e_3$ are squares in k.

- **(I9) Kubert's normal form** [Ku, Table 3, the entry for $\mathbb{Z}/10$, with $f = t$]. For t in a field k of characteristic 0 put

$$c = -\frac{t(t-1)(2t-1)}{t^2-3t+1}, \qquad b = \frac{t^3(t-1)(2t-1)}{(t^2-3t+1)^2}, \qquad E_t : \; y^2+(1-c)xy-by = x^3-bx^2.$$

  If E_t is an elliptic curve, $(0, 0)$ has order 10, and every pair (E', P) of an elliptic curve over k and a point $P \in E'(k)$ of order 10 is isomorphic over k to $(E_t, (0, 0))$ for some such $t \in k$. Kubert works over $\mathbb{Q}$, but his derivation from the Tate normal form $y^2 + (1 - c)xy - by = x^3 - bx^2$ with $P = (0, 0)$ uses only field operations and is valid over every field of characteristic 0; the parameter is recovered as $t = c^2/(b - c)$.

2 The CM case: a point count at 3

Lemma 1. No abelian surface over $\mathbb{F}_3$ has exactly 40 points.

Proof. Let B be an abelian surface over $\mathbb{F}_3$. By Weil, the characteristic polynomial of Frobenius is $(x^2-\beta_1 x+3)(x^2-\beta_2 x+3)$ with β_1, β_2 real, $|\beta_i| \le 2\sqrt{3}$, and $h(y) = (y-\beta_1)(y-\beta_2) = y^2-sy+t$ has integer coefficients. Then $\#B(\mathbb{F}_3) = (4-\beta_1)(4-\beta_2) = h(4) = 16-4s+t$. If $h(4) = 40$, then $t = 24+4s$. The roots are real, so $s^2 \ge 4t = 96+16s$, that is $(s-8)^2 \ge 160$, so $s \le -5$ or $s \ge 21$; and $|s| = |\beta_1+\beta_2| \le 4\sqrt{3} < 7$. Hence $s \in \{-5, -6\}$ and $h = (y+1)(y+4)$ or $h = y(y + 6)$, with the root -4 or -6 outside $[-2\sqrt{3}, 2\sqrt{3}]$. □

The certificate (Section 8, part A) lists all values of $h(4)$; the largest is 49.

The CM case. Let A be of type (CM) with a point $P_5 \in A(\mathbb{Q})$ of order 5, and suppose $A(\mathbb{Q})[2] \cong (\mathbb{Z}/2)^r$. As in [P2214, Section 3], A has potentially good reduction at 3, and by (I1)(c) good reduction $\widetilde{A}$ over a finite extension of $\mathbb{Q}_3$ with residue field $\mathbb{F}_3$. The subgroup of $A(\mathbb{Q})$ generated by P_5 and $A(\mathbb{Q})[2]$ has order $5 \cdot 2^r$, prime to 3, so by (I1)(b) $5 \cdot 2^r$ divides $\#\widetilde{A}(\mathbb{F}_3)$, which is at most $(1 + \sqrt{3})^4 < 56$ by (I1)(d). If $r \ge 3$, this forces $\#\widetilde{A}(\mathbb{F}_3) = 40$, contradicting Lemma 1. So $r \le 2$. This case uses neither simplicity over $\mathbb{Q}$ nor a polarization.

3 Type (S): the endomorphism field is cyclic of order 2 or 4

Proposition 2. Let A be of type (S) and simple over $\mathbb{Q}$, with $A(\mathbb{Q})[5] \neq 0$. Then Γ is cyclic of order 2 or 4.

Proof. We follow [P63, Section 7] with the prime 7 replaced by 5, using (I5) with $p = 5$. The field $\mathbb{Q}(\zeta_5)$ is cyclic of degree 4 over $\mathbb{Q}$, so if $\mathbb{Q}(\zeta_5) \subseteq L$, then Γ has a quotient isomorphic to C_4. Modulo 5 the discriminants -3, -1, $-\frac{1}{3}$ of the polynomials $X^2 - X + \frac{1}{c}$ for rotations of order 3, 4, 6 are 2, $4 = 2^2$ and 3; the nonzero squares modulo 5 are 1 and 4. By (I2) and simplicity, $\Gamma \neq C_1$.

1. $\Gamma \in \{C_3, C_6, D_6, D_{12}\}$. Let r be a rotation of order 3 or 6 generating the rotation subgroup and R its trace-one representative, with $c \in \{1, 3\}$ a 5-adic unit. Every element of Γ maps R to R or $1 - R$ [P63, Section 7], so $\mathcal{B} = \mathbb{Z}_5[R]$ is a $\mathbb{Z}_5$-order preserved by Γ. Its discriminant is a non-square unit modulo 5, so $\mathcal{B}$ is the ring of integers of the unramified quadratic extension of $\mathbb{Q}_5$; $R \in \mathcal{B}^\times$ because $\det R = 1/c$ is a unit, and its reduction is a root of an irreducible quadratic over $\mathbb{F}_5$, hence not in $\mathbb{F}_5$. By [P63, Lemma 6], $\mathbb{Q}(\zeta_5) \subseteq L$. But C_3, C_6 and the abelianizations C_2 and $C_2 \times C_2$ of D_6 and D_{12} have no quotient C_4.

2. $\Gamma = D_8$. Let r be a rotation of order 4 and R its trace-one representative, $R^2 - R + \frac{1}{2} = 0$. Again $\mathcal{B} = \mathbb{Z}_5[R]$ is preserved by Γ. Its discriminant $-1 \equiv 2^2$ is a nonzero square modulo 5, so by Hensel's lemma the roots λ_1, λ_2 of $X^2 - X + \frac{1}{2}$ lie in $\mathbb{Z}_5^\times$ with $\lambda_1 - \lambda_2$ a unit, and $\mathcal{B} \cong \mathbb{Z}_5 \times \mathbb{Z}_5$ with idempotents $e_1 = (R - \lambda_2)/(\lambda_1 - \lambda_2)$ and $e_2 = (R - \lambda_1)/(\lambda_2 - \lambda_1)$. The rotations fix R, hence e_1 and e_2; the reflections map R to $1 - R$ and, since $\lambda_1 + \lambda_2 = 1$, exchange e_1 and e_2. The stabilizer of e_1 is the rotation subgroup, and for $X \in e_1(D \otimes \mathbb{Q}_5)e_2$ we have $RXR^{-1} = (\lambda_1/\lambda_2)X$, so the character η of [P63, Lemma 7] maps a lift of r to λ_1/λ_2, the ratio of the eigenvalues of R, a primitive fourth root of unity. As $5 \nmid 4$, its reduction $\bar{\eta}$ has order 4. By [P63, Lemma 7], $\mathbb{Q}(\zeta_5) \subseteq L$; but the abelianization of D_8 is $C_2 \times C_2$.

3. $\Gamma = V_4$. The argument of [P63, Section 7, case V_4] applies verbatim with 7 replaced by 5: if $a = R^2$ (for a suitable r and trace-zero representative R with $\det R \in \mathbb{Z}_5^\times$) is not a square modulo 5, [P63, Lemma 6] applies; if it is, a reflection exchanges the idempotents $(1 \pm R/b)/2$, $\bar{\eta}$ is the nontrivial quadratic character, and [P63, Lemma 7] applies. In both cases $\mathbb{Q}(\zeta_5) \subseteq L$, which is impossible because $[L : \mathbb{Q}] = 4$ and V_4 is not cyclic.

The remaining groups are C_2 and C_4. □

4 Type (S) with a cyclic quartic endomorphism field

Proof of Proposition A. Here A is of type (S) and $\Gamma = \langle r \rangle \cong C_4$. By (I6) with $n = 4$, $F = \mathrm{End}^0_{\mathbb{Q}}(A) \cong \mathbb{Q}(i)$, $G_{\mathbb{Q}}$ acts $F \otimes \mathbb{Q}_\ell$-linearly on the free rank-2 module $V_\ell(A)$, and $\varepsilon(r) = R/\bar{R} \in \{i, -i\}$. Take $\ell = 2$. Then $F \otimes \mathbb{Q}_2 = \mathbb{Q}_2(i)$ is a field with ring of integers $\mathbb{Z}_2[i]$ and uniformizer $\pi = 1 + i$.

Let p be an odd prime of good reduction for A, unramified in L, whose arithmetic Frobenius $g = \mathrm{Frob}_p \in G_{\mathbb{Q}}$ satisfies $\varepsilon(g) = \pm i$. By (I6), g acts as $B \otimes \gamma$ with $\gamma \in F^\times$ and $\gamma^2 = \varepsilon(g)\,\delta$, $\delta = \deg(u_g) > 0$. Writing $\gamma = x + yi$ with $x, y \in \mathbb{Q}$, the equations $x^2 - y^2 = 0$ and $2xy = \pm\delta$ give $\gamma = x(1 \pm i)$. Hence the trace of g over $\mathbb{Q}_2(i)$ is

$$a = \mathrm{tr}(B)\,\gamma \in (1 \pm i)\,\mathbb{Q}_2.$$

The roots of the characteristic polynomial $X^2 - aX + b$ of g over $\mathbb{Q}_2(i)$ are eigenvalues of Frobenius, hence algebraic integers, so $a \in \mathbb{Z}_2[i]$; writing $a = (1 \pm i)u$ with $u \in \mathbb{Q}_2$, the valuation $v_\pi(a) = 1 + 2v_2(u) \geq 0$ gives $u \in \mathbb{Z}_2$. By (I6), $b = \chi_2(g)\varepsilon(g) = \pm ip$. With (I1),

$$\#\widetilde{A}(\mathbb{F}_p) = \det_{\mathbb{Q}_2}(1 - g) = N_{\mathbb{Q}_2(i)/\mathbb{Q}_2}(1 - a + b) = N\big((1 - u) \pm i(p - u)\big) = (1 - u)^2 + (p - u)^2.$$

Suppose $p \equiv 3 \pmod 4$. If $u \in 2\mathbb{Z}_2$, both $1-u$ and $p-u$ are units and the sum is $\equiv 2 \pmod 8$. If $u \in 1+2\mathbb{Z}_2$, write $1-u=2v$ and $p-u=2w$; then $w-v=(p-1)/2$ is odd, so v^2+w^2 is a unit and the sum is $\equiv 4 \pmod 8$. In both cases $8 \nmid \#\widetilde{A}(\mathbb{F}_p)$.

There are infinitely many such primes. If $\mathbb{Q}(i) \subseteq L$, then $\mathbb{Q}(i)$ is the quadratic subfield of L, and every prime whose Frobenius generates $\mathrm{Gal}(L/\mathbb{Q})$ is inert in $\mathbb{Q}(i)$, that is $p \equiv 3 \pmod 4$. (This case does not occur: the argument of [P63, Section 9.1], which uses only that every element of Γ is the class of a power R^k with $\det R^k > 0$, shows that L is totally real. We do not need this.) Otherwise $L \cap \mathbb{Q}(i) = \mathbb{Q}$, $\mathrm{Gal}(L(i)/\mathbb{Q}) \cong \mathrm{Gal}(L/\mathbb{Q}) \times \mathrm{Gal}(\mathbb{Q}(i)/\mathbb{Q})$, and Chebotarev's theorem gives infinitely many primes whose Frobenius generates $\mathrm{Gal}(L/\mathbb{Q})$ and which are $\equiv 3 \pmod 4$. For each such p, $A(\mathbb{Q})[2^\infty]$ injects into $\widetilde{A}(\mathbb{F}_p)$ by (I1)(b), so its order divides 4. □

Proposition A uses no point of order 5, no simplicity over $\mathbb{Q}$ (a surface with $\Gamma \cong C_4$ is simple over $\mathbb{Q}$, because $\mathrm{End}^0_{\mathbb{Q}}(A)$ is a field) and no polarization.

5 The quadratic case

In this section A is simple over $\mathbb{Q}$, geometrically split, and either of type (N) or of type (S) with $\Gamma \cong C_2$. In type (N) let K be the quadratic field over which the two elliptic curves of $A_{\bar{\mathbb{Q}}}$ are defined (they are exchanged by σ); in type (S) put $K=L$. Every elliptic curve $E \subset A_{\bar{\mathbb{Q}}}$ that we consider is defined over K, and $E^\sigma \neq E$, since otherwise E would be stable under $G_{\mathbb{Q}}$ and defined over $\mathbb{Q}$ [P2214, Section 2], contradicting the simplicity of A over $\mathbb{Q}$.

Lemma 4 (a curve with full 2-torsion). Suppose that A is principally polarized and that $A(\mathbb{Q})[2]$ has rank at least 3. Then there is an elliptic curve $E_1 \subset A_K$, defined over K, with $E_1^\sigma \neq E_1$ and $E_1[2] \subseteq E_1(K)$.

Proof. Let λ be the principal polarization and e the Weil pairing on $A[2]$ attached to it, $e(z,z') = e_2^A(z, \lambda z')$, where $e_2^A : A[2] \times A^\vee[2] \to \mu_2$ is the canonical pairing; e is a nondegenerate alternating $G_{\mathbb{Q}}$-invariant pairing. If $A(\mathbb{Q})[2] = A[2]$, any elliptic curve $E \subset A_K$ defined over K works. So let $W = A(\mathbb{Q})[2]$ have dimension 3. Each $g \in G_{\mathbb{Q}}$ acts trivially on W and on the line $A[2]/W$, so $g = 1 + N_g$ with $N_g(A[2]) \subseteq W$ and $N_g(W) = 0$. For $w \in W$, $e(x + N_g x, w) = e(x,w)$ gives $e(N_g x, w) = 1$, so $N_g(A[2])$ lies in the line $W^\perp$. Hence N_g is one of the two maps $A[2]/W \to W^\perp$, and the image of $G_{\mathbb{Q}}$ in $\mathrm{GL}(A[2])$ has order at most 2.

Let $E \subset A_K$ be an elliptic curve defined over K.

1. $E[2] \neq E^\sigma[2]$ *as subgroups of* A. Every lift $\tilde{\sigma}$ maps $E[2]$ onto $E^\sigma[2] \neq E[2]$, so it acts nontrivially on $A[2]$, as the unique nontrivial element u of the image. If some $h \in G_K$ acted as u, the lift $\tilde{\sigma}h$ would act trivially. So G_K acts trivially on $A[2]$, and $E[2] \subseteq E(K)$. Take $E_1 = E$.

2. $E[2] = E^\sigma[2] =: \Lambda$. The subgroup Λ is stable under G_K and under $\tilde{\sigma}$, hence under $G_{\mathbb{Q}}$. Write $\iota : E \to A_K$ for the inclusion, $e_2^E : E[2] \times E^\vee[2] \to \mu_2$ for the canonical pairing and $\phi_E : E \to E^\vee$ for the principal polarization, so that $e_E(x,y) = e_2^E(x, \phi_E y)$ is the Weil pairing of E. By functoriality, $e(\iota x, z) = e_2^E(x, \iota^\vee \lambda z)$ for $x \in E[2]$ and $z \in A[2]$. The polarization $\iota^\vee \lambda \iota$ of E is $n\phi_E$ for some $n \geq 1$, so $e(\iota x, \iota y) = e_E(x,y)^n$ for $x, y \in E[2]$.

   If n is even, Λ is isotropic, hence $\Lambda = \Lambda^\perp$, and $x \mapsto e(x, \cdot)$ identifies $A[2]/\Lambda$ with the dual $\Lambda^\vee$ as $G_{\mathbb{Q}}$-modules. If $\Lambda \not\subseteq W$, then $W \cap \Lambda$ is a line and $W + \Lambda = A[2]$, so $G_{\mathbb{Q}}$ acts trivially on $A[2]/\Lambda \cong \Lambda^\vee$, hence on Λ, and $\Lambda \subseteq W$ after all. So $E[2] = \Lambda \subseteq A(\mathbb{Q})$, and $E[2] \subseteq E(K)$. Take $E_1 = E$.

If n is odd, e is nondegenerate on Λ, so $A[2] = \Lambda \oplus \Lambda^\perp$ as $G_\mathbb{Q}$-modules and $W = \Lambda^{G_\mathbb{Q}} \oplus (\Lambda^\perp)^{G_\mathbb{Q}}$. As $\dim W = 3$, either $\Lambda \subseteq W$, and we take $E_1 = E$, or $\Lambda^\perp \subseteq W$. In the latter case let $E^\perp$ be the complementary elliptic curve, the connected component of the kernel of $\iota^\vee\lambda : A_K \to E^\vee$; it is defined over K. For $x \in E[2]$ and $y \in E^\perp[2]$ we have $e(\iota x, y) = e_2^E(x, \iota^\vee\lambda y) = 1$, so $E^\perp[2] \subseteq \Lambda^\perp$. Both groups have order 4, so $E^\perp[2] = \Lambda^\perp \subseteq A(\mathbb{Q})$, and $E^\perp[2] \subseteq E^\perp(K)$. Finally $(E^\perp)^\sigma \neq E^\perp$: otherwise $E^\perp$ would be stable under $G_\mathbb{Q}$, hence defined over $\mathbb{Q}$, contradicting the simplicity of A over $\mathbb{Q}$. Take $E_1 = E^\perp$.

In all cases E_1 has the required properties. □

Lemma 5 (Weil restriction). Let $E \subset A_K$ be an elliptic curve defined over K with $E^\sigma \neq E$.

(a) A is isogenous over $\mathbb{Q}$ to $\mathrm{Res}_{K/\mathbb{Q}} E'$ for every elliptic curve E' over K that is K-isogenous to E.

(b) Let p be an odd prime, inert in K, at which A has good reduction, and put $\mathfrak{p} = pO_K$. Then every such E' has good reduction at $\mathfrak{p}$, and $\#\widetilde{A}(\mathbb{F}_p) = \#\widetilde{E'}(O_K/\mathfrak{p})$, where $O_K/\mathfrak{p} \cong \mathbb{F}_{p^2}$.

(c) If $A(\mathbb{Q})$ has a point of order 5 and $E[2] \subseteq E(K)$, then some E' that is K-isogenous to E has $E'(K) \supseteq \mathbb{Z}/2 \times \mathbb{Z}/10$.

Proof. (a) Let $\iota : E \to A_K$ be the inclusion and $\iota^\sigma : E^\sigma \to A_K$ its conjugate. The map $(x, y) \mapsto \iota(x) + \iota^\sigma(y)$ from $(\mathrm{Res}_{K/\mathbb{Q}} E)(\bar{\mathbb{Q}})$ to $A(\bar{\mathbb{Q}})$ is a homomorphism of abelian varieties over $\bar{\mathbb{Q}}$, and it commutes with the action of $G_\mathbb{Q}$ described in (I7), because $g \circ \iota = \iota^g \circ g$ for $g \in G_\mathbb{Q}$. By Galois descent it is a homomorphism $\mathrm{Res}_{K/\mathbb{Q}} E \to A$ over $\mathbb{Q}$. It is surjective because $E + E^\sigma = A_{\bar{\mathbb{Q}}}$, hence an isogeny. A K-isogeny $E' \to E$ induces a $\mathbb{Q}$-isogeny $\mathrm{Res}_{K/\mathbb{Q}} E' \to \mathrm{Res}_{K/\mathbb{Q}} E$.

(b) Fix a prime $\ell \neq p$. By (a) and (I7), $V_\ell(A) \cong \mathrm{Ind}_{G_K}^{G_\mathbb{Q}} V_\ell(E')$ as $G_\mathbb{Q}$-modules. By (I1)(a) this is unramified at p. As p is unramified in K, the inertia groups at primes above p lie in G_K, and $V_\ell(E')$ is a G_K-submodule of $V_\ell(A)$; so it is unramified at $\mathfrak{p}$, and E' has good reduction at $\mathfrak{p}$. Let $g \in G_\mathbb{Q}$ be an arithmetic Frobenius element at a prime above p, well defined up to conjugation and inertia, which acts trivially. It does not lie in G_K, as p is inert, and $g^2 \in G_K$ is an arithmetic Frobenius element at $\mathfrak{p}$, because $O_K/\mathfrak{p}$ has degree 2 over $\mathbb{F}_p$. In a decomposition $\mathrm{Ind}\, V = V \oplus gV$, g has the block form $\left(\begin{smallmatrix} 0 & B \\ C & 0 \end{smallmatrix}\right)$, so $\det(1 - g) = \det(1 - BC)$, where BC is the action of g^2 on $V_\ell(E')$. By (I1), $\#\widetilde{A}(\mathbb{F}_p) = \det(1 - g \mid V_\ell(A)) = \det(1 - g^2 \mid V_\ell(E')) = \#\widetilde{E'}(O_K/\mathfrak{p})$.

(c) Let $P_5 \in A(\mathbb{Q})$ have order 5 and let $\pi : A_K \to Y = A_K/E^\sigma$ be the quotient map; its restriction to E is a K-isogeny $E \to Y$. By (I3) over K, either $P_5 \in E^\sigma$, and then $P_5 = \tilde{\sigma}(P_5) \in E(K)$, or $\pi(P_5) \in Y(K)$ has order 5. So there is a K-isogeny $f : E \to E_5$ with $E_5 \in \{E, Y\}$ and a point of order 5 in $E_5(K)$. Let H be the 5-primary part of $\ker f$, $a : E \to E' = E/H$ and $b : E' \to E_5$ with $f = b \circ a$. As $\deg a$ is a power of 5, a maps $E[2]$ isomorphically and G_K-equivariantly onto $E'[2]$, so $E'[2] \subseteq E'(K)$. As $\deg b$ is prime to 5, the dual isogeny $\hat{b}$ is injective on $E_5[5]$ and maps the point of order 5 to a point of order 5 in $E'(K)$, as in the proof of [P63, Lemma 1]. □

Lemma 6 (a Chebotarev criterion). Let $K = \mathbb{Q}(\sqrt{d})$ be a quadratic field and $E : y^2 = (x - e_1)(x - e_2)(x - e_3)$ an elliptic curve over K with $e_1, e_2, e_3 \in K$, and put $N_{ij} = N_{K/\mathbb{Q}}(e_i - e_j) \in \mathbb{Q}^\times$. Suppose that 8 divides $\#\widetilde{E}(O_K/pO_K)$ for all but finitely many primes p inert in K. Then, for a suitable numbering $\{i, j, k\} = \{1, 2, 3\}$,

$$N_{ij} \in \mathbb{Q}^{\times 2} \qquad \text{and} \qquad N_{ik} \in \mathbb{Q}^{\times 2} \ \text{ or } \ N_{jk} \in \mathbb{Q}^{\times 2} \ \text{ or } \ d\,N_{ik}N_{jk} \in \mathbb{Q}^{\times 2}.$$

Proof. For all but finitely many inert p, we have $p > 3$, the e_i are integral at $\mathfrak{p} = pO_K$ with unit differences, and the reduction $\widetilde{E}$ over $k = O_K/\mathfrak{p} \cong \mathbb{F}_{p^2}$ has full rational 2-torsion. Then $8 \mid \#\widetilde{E}(k)$ if and only if $\widetilde{E}(k)$ contains a point of order 4, that is, some $(\bar{e}_m, 0)$ lies in $2\widetilde{E}(k)$; by (I8) this means that $\bar{e}_m - \bar{e}_{m'}$ is a square in k for both $m' \neq m$. An element of $\mathbb{F}_{p^2}^\times$ is a square if and only if its norm to $\mathbb{F}_p$ is, and since σ induces the Frobenius of $k/\mathbb{F}_p$, the norm of $\bar{e}_m - \bar{e}_{m'}$ is $N_{mm'}$ mod p. So for all but finitely many inert p:

$$(*) \qquad \text{for some } m,\ \Big(\frac{N_{mm'}}{p}\Big) = 1 \text{ for both } m' \neq m.$$

Let V be the subgroup of $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ generated by the classes of d, N_{12}, N_{13}, N_{23}, written additively as an $\mathbb{F}_2$-vector space. By Chebotarev's theorem for $\mathbb{Q}(\sqrt{d}, \sqrt{N_{12}}, \sqrt{N_{13}}, \sqrt{N_{23}})$, every homomorphism $\phi : V \to \mathbb{F}_2$ is $x \mapsto \big(1 - (x/p)\big)/2$ for infinitely many primes p, and these p are inert in K when $\phi(d) = 1$. So $(*)$ says: no ϕ with $\phi(d) = 1$ takes the value 1 on two of N_{12}, N_{13}, N_{23}. A system $\phi(v_1) = \cdots = \phi(v_s) = 1$ is solvable unless some sum of an odd number of the v_t vanishes in V. For each pair $\{x, y\} \subset \{N_{12}, N_{13}, N_{23}\}$ the system $\phi(d) = \phi(x) = \phi(y) = 1$ is therefore unsolvable, so $x = 0$, $y = 0$ or $x + y = d$ in V ($d \neq 0$ as $K \neq \mathbb{Q}$). If none of the three classes were 0, then $N_{12} + N_{13} = N_{12} + N_{23} = N_{13} + N_{23} = d$, so all three would be equal and $d = 0$. Hence some N_{ij} is a square, and the pair $\{N_{ik}, N_{jk}\}$ gives the second condition. □

The condition is also sufficient: if it holds, then $(*)$ holds at every inert p as in the proof (if $dN_{ik}N_{jk}$ is a square, exactly one of N_{ik}, N_{jk} is a square modulo p), so 8 divides $\#\widetilde{E}(O_K/pO_K)$. The curve $y^2 = x(x-1)(x-\sqrt{2})$ over $\mathbb{Q}(\sqrt{2})$, with $N_{12} = 1$, $N_{13} = -2$, $N_{23} = -1$ and $dN_{13}N_{23} = 4$, shows that the third alternative is needed.

Let $\mathcal{X}$ be the elliptic curve $y^2 = x^3 - 2x^2 + 1$ over $\mathbb{Q}$, with origin the point at infinity O (Cremona 20a2). Put

$$e_1 = -\frac{t^3(t-1)}{t^2-3t+1}, \quad B_t = \frac{-8t^4+12t^3-4t+1}{(t^2-3t+1)^2}, \quad \rho = \frac{(2t-1)^2 s}{(t^2-3t+1)^2}, \quad e_{2,3} = \frac{-B_t \pm \rho}{8}.$$

Lemma 7 ($\mathbb{Z}/2 \times \mathbb{Z}/10$ over quadratic fields).

(a) $\mathcal{X}$ has rank 0 and $\mathcal{X}(\mathbb{Q}) = \{O, (0, \pm 1), (1, 0), (2, \pm 1)\}$.

(b) Let K be a quadratic field and E' an elliptic curve over K with $E'(K) \supseteq \mathbb{Z}/2 \times \mathbb{Z}/10$. Then $E' \cong E_t$ over K for some $t \in K$ such that $s^2 = (2t-1)(4t^2-2t-1)$ for some $s \in K$, and E' is isomorphic over K to $y^2 = (x-e_1)(x-e_2)(x-e_3)$. The point $Q = (2t, s) \in \mathcal{X}(K)$ does not lie in $\mathcal{X}(\mathbb{Q})$.

(c) Put $R = Q + Q^\sigma \in \mathcal{X}(\mathbb{Q})$. If $R = O$, then $t \in \mathbb{Q}$. If $R \neq O$, then the line through Q and Q^σ passes through $-R$ and has a slope $m \in \mathbb{Q}$; the x-coordinates of Q and Q^σ are the roots of the monic quadratic

$$q_{R,m}(X) = \frac{X^3 - 2X^2 + 1 - \big(y(-R) + m(X - x(-R))\big)^2}{X - x(-R)} \in \mathbb{Q}[X],$$

which is irreducible over $\mathbb{Q}$, and $K = \mathbb{Q}\big(\sqrt{\operatorname{disc} q_{R,m}}\big)$.

Proof. (a) is part B of the certificate (Magma proves the rank is 0; the torsion is $\mathbb{Z}/6$). (b) By (I9), $(E', P) \cong (E_t, (0,0))$ for a point P of order 10; as E_t is an elliptic curve, $t \neq \frac{1}{2}$ and $t^2 - 3t + 1 \neq 0$. The certificate (part C) checks that $e_1 = x(5 \cdot (0,0))$ is a root of the

2-division polynomial $4x^3 + b_2x^2 + 2b_4x + b_6$ of E_t, that the complementary quadratic factor is $4x^2 + B_tx + C_t$ with

$$B_t^2 - 16C_t = \frac{(2t-1)^5(4t^2-2t-1)}{(t^2-3t+1)^4} = (2t-1)(4t^2-2t-1)\cdot\Big(\frac{(2t-1)^2}{(t^2-3t+1)^2}\Big)^2,$$

and that over $\mathbb{Q}(t)(s)$, $s^2 = (2t-1)(4t^2-2t-1)$, the 2-division polynomial is $4\prod(x-e_i)$. Since $E'[2] \subseteq E'(K)$, this discriminant is a square in K; the factor $(2t-1)^2/(t^2-3t+1)^2$ is nonzero, so $(2t-1)(4t^2-2t-1) = s^2$ with $s \in K$, and completing the square identifies E_t with $y^2 = \prod(x-e_i)$. The six points of $\mathcal{X}(\mathbb{Q})$ have $t = x/2 \in \{\infty, 0, \frac{1}{2}, 1\}$, where E_t is not defined or is singular (certificate, part B); also no elliptic curve over $\mathbb{Q}$ has a subgroup $\mathbb{Z}/2 \times \mathbb{Z}/10$ [Ma]. (c) The point R is fixed by σ, so $R \in \mathcal{X}(\mathbb{Q})$. If $R = O$, then $Q^\sigma = -Q$ has the same x-coordinate, so $x(Q) \in \mathbb{Q}$. If $R \neq O$, then $Q \neq Q^\sigma$ and the line through them is σ-stable, hence defined over $\mathbb{Q}$; its third intersection with $\mathcal{X}$ is $-(Q+Q^\sigma) = -R$. It is not vertical, since the vertical line through $-R$ meets $\mathcal{X}$ only in $\pm R$ and O. The remaining statements follow. □

Proposition 8. Let $R \in \mathcal{X}(\mathbb{Q}) \setminus \{O\}$ and $m \in \mathbb{Q}$ be such that $q_{R,m}$ is irreducible over $\mathbb{Q}$. Let x_Q be a root of $q_{R,m}$, $K = \mathbb{Q}(x_Q)$, $t = x_Q/2$ and $s = y(-R) + m(x_Q - x(-R))$, so that $Q = (x_Q, s) \in \mathcal{X}(K)$, and suppose that E_t is an elliptic curve. Then the conclusion of Lemma 6 fails for $y^2 = (x-e_1)(x-e_2)(x-e_3)$ over K, with e_1, e_2, e_3 as above.

Proof. The other root of $q_{R,m}$ gives Q^σ and the conjugates of the e_i, with the same norms, so the choice of x_Q does not matter. Over the function field $\mathbb{Q}(m)$, let $K_m = \mathbb{Q}(m)[X]/(q_{R,m})$, $t = X/2$, $s = y(-R) + m(X - x(-R))$; then $s^2 = (2t-1)(4t^2-2t-1)$, and the norms $N_{ij} \in \mathbb{Q}(m)$ of $e_i - e_j$ and $d = \operatorname{disc} q_{R,m}$ are computed in part D of the certificate. Up to squares in $\mathbb{Q}(m)^\times$, in each of the five families two of the three norms have a common class g_R, and the third is -1 or $-d$:

R	$d = \operatorname{disc} q_{R,m}$	g_R	third norm
$(0,1)$	$m^4 + 4m^2 - 8m + 4$	$(m-1)(m^2+m-1)$	-1
$(0,-1)$	$m^4 + 4m^2 + 8m + 4$	$-(m+1)(m^2-m-1)$	-1
$(2,-1)$	$m^4 - 8m^2 + 8m$	$-(m-1)(m^2-3m+1)$	$-d$
$(2,1)$	$m^4 - 8m^2 - 8m$	$(m+1)(m^2+3m+1)$	$-d$
$(1,0)$	$m^4 - 2m^2 + 5$	$(m^2-1)(m^2-5)$	-1

Let $m_0 \in \mathbb{Q}$ be as in the statement, and suppose first that m_0 is not a root of any numerator or denominator of d, of the N_{ij}, or of $N_{K_m/\mathbb{Q}(m)}(t^2-3t+1)$. Let $\mathcal{L} = \mathbb{Q}[m]_{(m-m_0)}$ be the local ring of $\mathbb{Q}[m]$ at m_0. As $q_{R,m}$ is monic in X with coefficients in $\mathbb{Q}[m]$, the ring $\mathcal{O} = \mathcal{L}[X]/(q_{R,m})$ is free over $\mathcal{L}$ with basis $1, X$, and $\mathcal{O}/(m-m_0)\mathcal{O} \cong \mathbb{Q}[X]/(q_{R,m_0}) \cong K$, with $X \mapsto x_Q$. Norms from $\mathcal{O}$ to $\mathcal{L}$ are determinants of multiplication maps in this basis, so they commute with reduction modulo $m - m_0$. The element $t^2 - 3t + 1$ of $\mathcal{O}$ has norm $N_{K_m/\mathbb{Q}(m)}(t^2-3t+1) \in \mathbb{Q}[m]$, which does not vanish at m_0, so it is a unit of $\mathcal{O}$. The e_i have denominators dividing $8(t^2-3t+1)^2$, so they lie in $\mathcal{O}$, and their reductions are the e_i of the statement. Hence $N_{ij}(m_0) = N_{K/\mathbb{Q}}(e_i - e_j)$, and $d(m_0) = \operatorname{disc} q_{R,m_0}$. Each identity of square classes in the table reads $N = \kappa h^2$ with $\kappa \in \{g_R, -1, -d\}$ and $h \in \mathbb{Q}(m)^\times$. The zeros and poles of h lie among those of N, g_R and d, and the roots of g_R are roots of numerators or denominators of the N_{ij}. So $h(m_0) \in \mathbb{Q}^\times$, $g_R(m_0) \neq 0$, and the classes above hold at m_0. The conclusion of Lemma 6 would need some norm to be a square. The class -1 is not a square. If the third norm is $-d$ and is a square, the second condition requires $g_R(m_0) \in \mathbb{Q}^{\times 2}$ or $d \cdot g_R(m_0)^2 \in \mathbb{Q}^{\times 2}$; the latter would make both d and $-d$ squares. So in every case $g_R(m_0) \in \mathbb{Q}^{\times 2}$, and (m_0, y) is a rational point on $C_R : y^2 = g_R(m)$. The five curves C_R have genus 1 and Jacobians $40a3$ (for $R = (0, \pm 1), (1, 0)$) and $20a2$ (for $R = (2, \pm 1)$), of rank 0

(part E of the certificate). Their rational points are the points at infinity and those with $m \in \{0,1\}$, $\{-1,0\}$, $\{0,1,2\}$, $\{-2,-1,0\}$, $\{\pm 1\}$ for $R = (0,1),(0,-1),(2,-1),(2,1),(1,0)$ respectively; for each of these m, $q_{R,m}$ splits over $\mathbb{Q}$. Finally, part F of the certificate treats the excluded values m_0 directly: each gives a reducible q_{R,m_0}, a value of t with $t^2-3t+1=0$, or a singular E_t. □

The quadratic case. Let A be principally polarized, simple over $\mathbb{Q}$, of type (N) or of type (S) with $\Gamma \cong C_2$, with a point of order 5 in $A(\mathbb{Q})$, and suppose $A(\mathbb{Q})[2]$ has rank at least 3. By Lemma 4 there is $E_1 \subset A_K$ over K with $E_1^\sigma \neq E_1$ and $E_1[2] \subseteq E_1(K)$. By Lemma 5(c) some E', K-isogenous to E_1, has $E'(K) \supseteq \mathbb{Z}/2 \times \mathbb{Z}/10$, and by Lemma 5(a) A is isogenous over $\mathbb{Q}$ to $\mathrm{Res}_{K/\mathbb{Q}} E'$. The subgroup of $A(\mathbb{Q})$ generated by the point of order 5 and $A(\mathbb{Q})[2]$ has order divisible by 40; by (I1)(b) and Lemma 5(b), $40 \mid \#\widetilde{A}(\mathbb{F}_p) = \#\widetilde{E'}(O_K/pO_K)$ for every odd prime $p \neq 5$ inert in K at which A has good reduction. Let $E' \cong E_t$ and $Q \in \mathcal{X}(K)$ be as in Lemma 7.

- If $Q + Q^\sigma = O$, then $t \in \mathbb{Q}$, and E' is the base change of the elliptic curve E_t over $\mathbb{Q}$. By (I7), $\mathrm{Res}_{K/\mathbb{Q}} E'$ is isogenous over $\mathbb{Q}$ to $E_t \times E_t^{(K)}$, so A is not simple over $\mathbb{Q}$. This is a contradiction.
- Otherwise $R = Q + Q^\sigma \neq O$, and by Lemma 7(c) the line through Q and Q^σ has a slope $m \in \mathbb{Q}$, $q_{R,m}$ is irreducible with root $x(Q)$, $K = \mathbb{Q}(x(Q))$, and $y(Q) = y(-R) + m(x(Q) - x(-R))$. So R, m and $x_Q = x(Q)$ satisfy the hypotheses of Proposition 8. The curve E' is isomorphic over K to $y^2 = \prod(x - e_i)$, so the two have isomorphic reductions at all but finitely many primes of K, and Lemma 6 applies to $y^2 = \prod(x - e_i)$. Its conclusion contradicts Proposition 8.

Hence $A(\mathbb{Q})[2]$ has rank at most 2.

6 Proof of the Theorem and consequences for the census

Proof of the Theorem. Let $A/\mathbb{Q}$ be principally polarized, simple over $\mathbb{Q}$ and geometrically split, with a rational point of order 5. By (I2), A is of type (N), (CM) or (S). In type (CM), $A(\mathbb{Q})[2]$ has rank at most 2 by Section 2. In type (S), $\Gamma \cong C_2$ or C_4 by Proposition 2; if $\Gamma \cong C_4$, then $\#A(\mathbb{Q})[2^\infty] \leq 4$ by Proposition A. In type (N) and in type (S) with $\Gamma \cong C_2$, the rank is at most 2 by Section 5. □

Proof of the Corollary. The Jacobian of a curve of genus 2 over $\mathbb{Q}$ is a principally polarized abelian surface, and each of $[2,2,10]$, $[2,2,20]$ and $[2,2,2,10]$ contains a point of order 5 and has 2-rank at least 3. For Jacobians split over $\mathbb{Q}$, the 2-rank obstruction shows that rational 2-rank 4 excludes rational points of order 5. □

On the census, the three groups become impossible in the class "geometrically split, simple over $\mathbb{Q}$". For $[2,2,2,10]$, already impossible in the class "split over $\mathbb{Q}$", the combined geometrically split cell becomes impossible ($\emptyset$). The group $[2,2,10]$ has examples split over $\mathbb{Q}$: the census records the curve 2.2.10.b and a certified infinite family of Jacobians split over $\mathbb{Q}$ with exact torsion group $[2,2,10]$ (see its census page and the family certificate). The class "split over $\mathbb{Q}$" of $[2,2,20]$ stays open. So the combined cells of $[2,2,10]$ and $[2,2,20]$ keep their grades.

7 Remarks

1. *The point of order* 5 *is needed* in types (N), (CM) and (S) with $\Gamma \in \{C_2, V_4, D_8\}$: the LMFDB curves 2560.a.819200.1, 4096.b.65536.1, 25600.f.512000.1, 8192.b.131072.1 and

12800.c.128000.1 are simple over $\mathbb{Q}$, geometrically split, with rational torsion subgroup $(\mathbb{Z}/2)^3$ and Sato–Tate groups $N(\mathrm{SU}(2)\times\mathrm{SU}(2))$, $J(C_2)$, $J(E_1)$, $J(E_2)$ and $J(E_4)$ (component groups C_2, D_2, C_2, D_2, D_4). For $\Gamma\cong C_4$ (Sato–Tate group E_4) no point of order 5 is needed. Consistently, the LMFDB curves with Sato–Tate group E_4 that are simple over $\mathbb{Q}$ have torsion [2], [2, 2] or [2, 10]; the last is $X_1(16)$ (256.a.512.1), whose endomorphism field is $\mathbb{Q}(\zeta_{16})^+$, and for the primes $p<200$ with $p\equiv 3,11 \pmod{16}$ one finds $\#J_1(16)(\mathbb{F}_p)\equiv 4 \pmod 8$ (for example $\#J_1(16)(\mathbb{F}_3)=20=2^2+4^2$, $u=-1$ in the proof of Proposition A).

2. *A check of Lemmas 4–6 on known examples (not used in the proof).* For the eight LMFDB curves with rational torsion subgroup $(\mathbb{Z}/2)^3$ that are simple over $\mathbb{Q}$ and split over a quadratic field K (2169.a.175689.1, 2457.b.199017.1, 2560.a.5120.1, 2560.a.819200.1, 4096.b.65536.1, 4352.b.278528.1, 8192.b.131072.1, 25600.f.512000.1), the supplementary script `qsimple_2210_validation.m` checks that $\#J(\mathbb{F}_p)=\#E(\mathbb{F}_{p^2})$ at all inert primes $p<400$ of good reduction for the LMFDB elliptic factor E over K (and the product formula at split primes), finds in the 2-isogeny class of E over K a curve with full K-rational 2-torsion, as Lemma 4 predicts, and checks that each such curve satisfies the conclusion of Lemma 6, as it must since $8\mid\#J(\mathbb{F}_p)$. The curve 4096.b.65536.1 has CM and 8192.b.131072.1 has $\Gamma\cong V_4$, so they are not of the kind treated in Section 5; but in the proofs of Lemmas 4 and 5 the type of A enters only through the simplicity of A over $\mathbb{Q}$ and an elliptic curve $E\subset A_K$ defined over K, and Lemma 6 concerns elliptic curves over K. The script also checks the example after Lemma 6: 8 divides $\#\widetilde{E}(\mathbb{F}_{p^2})$ at every inert prime $p<400$.

3. *The polarization.* Only Lemma 4 uses that the polarization is principal. Section 2, Proposition 2 and Proposition A hold for every abelian surface over $\mathbb{Q}$ of the stated types.

4. *Why the argument for order 7 does not apply.* The proof for [2, 2, 14] used that no elliptic curve over a quadratic field has a subgroup $\mathbb{Z}/28$ or $\mathbb{Z}/2\times\mathbb{Z}/14$ [P2214, Lemma 1]. Over quadratic fields $\mathbb{Z}/2\times\mathbb{Z}/10$ does occur, for infinitely many fields; the quadratic points of $X_1(2,10)$ are therefore controlled here by the Frobenius condition at inert primes.

8 Computations

The certificate `qsimple_2210_checks.m` (Magma V2.30-1, under a second) consists of assertions and ends with `ALL_CHECKS_PASSED`; its log is `qsimple_2210_checks.log`. SHA-256 of the script: `b5ee08ba8354b767816e762666e18aa2177c000fd03fd2b897f8d6d86f01bc05`; of the log: `29c3bc4a93d8034376ee575ce865fecbff3ebcbabce9e83470a1e0dd9ff070ee`. Part A lists the values $h(4)$ of Lemma 1. Part B checks that $\mathcal{X}$ is $20a2$ with rank 0 (proved by Magma) and torsion $\mathbb{Z}/6$, lists $\mathcal{X}(\mathbb{Q})$, and checks that E_t is singular at $t=0,\frac{1}{2},1$. Part C checks the Kubert family, the formula $t=c^2/(b-c)$, the discriminant in Lemma 7(b) and the formulas for e_1,e_2,e_3 over $\mathbb{Q}(t)(s)$. Part D computes, for each $R\neq O$, the quadratic $q_{R,m}$, the field K_m, the identity $s^2=(2t-1)(4t^2-2t-1)$, the identity $4x^3+b_2x^2+2b_4x+b_6=4\prod(x-e_i)$ for the specialized Kubert curve, the norms N_{ij} and their square classes, and asserts the table of Proposition 8, including the discriminants d. Part E proves that each C_R has rank 0, checks the Cremona label of its Jacobian, checks that the points found are all of $C_R(\mathbb{Q})$ (their number equals the order of the torsion of the Jacobian) and that their finite slopes are those listed in Proposition 8, and checks that each of these slopes gives a reducible $q_{R,m}$.

Part F computes the excluded slopes of Proposition 8 (rational roots of all numerators and denominators involved) and checks from scratch that each gives a reducible $q_{R,m}$, a value of t with $t^2 - 3t + 1 = 0$, or a singular E_t. Part G checks, at the sample slope $m = 7/3$ in each family, that $Q + Q^\sigma = R$, that $E_t(K)_{\mathrm{tors}} \cong \mathbb{Z}/2 \times \mathbb{Z}/10$, and that the conclusion of Lemma 6 fails there.

The supplementary script `qsimple_2210_validation.m` of Remark 2 (a few seconds) also consists of assertions and ends with `VAL_DONE`; its log is `qsimple_2210_validation.log`.

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